Animal Record Keeping (Short Note)

- "You cannot manage what you do not measure," so record keeping is essential for profitable livestock management
- Record keeping means organizing and storing information about farm activities and animals in a usable format
- Animal recording involves collecting and managing data about individual animals on the farm

Basic guidelines

- Each animal must be individually identified for meaningful records (ear tags are most practical)
- Record important events like birth, parents, calving date, sex, birth weight, and management practices (treatments, vaccination)
- Maintain backup and store records for at least 5 years

Criteria for good record keeping

- Must be useful for management decisions
- Simple and easy to interpret
- Avoid duplication
- Must be kept in a form that can be easily converted to information
- Should lead to action such as culling or nutritional correction

Importance / Objectives

- Evaluate performance for selection
- Economic evaluation and assess profit or loss
- Genetic evaluation for breeding
- Control inbreeding
- Aid in culling low performers
- Improve bargaining power on products
- Better decision-making in feeding, treatment, mating, weaning, and selling
- Aid in disease management and husbandry research

Types of records kept on the farm

- Physical identification records

- Breeding records
- Production or performance records
- Feeding records
- Financial records
- Health records
- Growth/weight and milk production records
- Lambing records and carcass yield, etc.

Types of Recording Systems in Livestock Development (Brief)

In livestock development, different recording systems are used to monitor animal identity, performance, health, and economic productivity. Major types include:

• Physical (Identification) Records

Keep information on animal ID, sex, breed, ancestry, and date of birth. Essential for traceability and correct data management.

• Breeding Records

Track heat dates, service dates, sire and dam ID, pregnancy status, calving dates, drying off, and remarks. Helps improve reproductive efficiency and maintain yearly calving.

• Production (Performance) Records

Assess traits like milk yield, growth rate, carcass characteristics, egg production, etc., used for genetic improvement and economic evaluation.

• Feeding Records

Document feed type, quantity, availability of fodder, supplements, minerals, and leftovers. Useful for ration planning and cost analysis.

• Health Records

Include vaccination, treatment, deworming, morbidity and mortality data. Help in disease prevention and herd health planning.

• Financial Records

Maintain income and expenditure details for profitability assessment, credit/loan access, and enterprise analysis.

• **Growth / Weight Records**

Record periodic body weights from birth to sale or weaning to evaluate growth performance.

• **Milk and Lactation Records**

Weekly milk yield, lactation length, milk used for calves or sold, are useful for dairy selection and management.

• **Lambing or Birth Records**

For small ruminants: birth date, dam ID, birth weight, type of birth (single/twins), sex.

• **Carcass and Skin Quality Records**

Useful in meat and hide production systems to improve meat yield and skin traits.

Conservation of Animal Genetic Resources (Short Note)

- Animal genetic resources include all breeds and populations of livestock that contribute to food, agriculture, and rural livelihoods.
- Conservation is needed because indiscriminate crossbreeding, replacement of indigenous breeds, and environmental changes are causing many native breeds to become endangered.
- Indigenous breeds are highly valuable since they are adapted to local climate, disease, and low-cost management. Their loss reduces genetic variability and future improvement potential.

Types of conservation methods

• **In situ conservation**: Maintaining breeds in their natural environment through farm-level breeding programs and controlled management.

• **Ex situ conservation**: Preserving genetic material outside the natural habitat, such as semen, embryos, and DNA in cryogenic gene banks.

Purpose/Importance

- Ensure long-term genetic diversity
- Help future breeding and research programs
- Support sustainable livestock production under changing environments

Examples

• Breed improvement programs and genetic evaluation support conservation of valuable indigenous livestock.

Point	**In situ Conservation**	**Ex situ Conservation**
Definition	Maintenance of living populations of animals in their original habitats.	Conservation of plants or animal genetic resources outside their natural habits.
Appropriation	Best way to conserving bio diversity.	Not
Infrastructure	Not	Requires appropriate infrastructure organization, technical capacity, legal arrangement etc.
Level	Only species level.	Protect diversity at both species and genetic level
Rare species	Not suitable for conserving rare species or outside protected area.	Suitable
Division	Haven't such division.	It is divided into ex situ in vivo & ex situ in vitro.

Short Note on Animal Breeding Program

An **animal breeding program** is a **planned and systematic process** designed to **improve the genetic quality and performance** of farm animals through **selection, mating, and evaluation** over generations.

Objectives:

1. To **increase productivity** (milk, meat, eggs, wool).
2. To **improve fertility, growth rate, and disease resistance**.
3. To **develop superior breeds or strains** suited to local environments.
4. To **combine desirable traits** from different breeds through crossbreeding.

Main Steps:

1. **Setting Breeding Goals:** Define traits to improve (e.g., milk yield, fertility).
2. **Selection:** Identify and choose genetically superior animals using performance, pedigree, or progeny records.
3. **Mating Plan:** Use suitable mating systems — inbreeding, line breeding, or crossbreeding — depending on objectives.

4. **Performance Recording:** Maintain detailed records to evaluate genetic progress.
5. **Evaluation and Culling:** Assess offspring performance and remove inferior animals.

Types of Breeding Programs:

- **Pure Breeding:** Within the same breed to maintain and improve it.
- **Crossbreeding:** Between different breeds to combine superior traits and produce hybrid vigor.
- **Grading-up:** Using purebred sires on non-descript females for gradual improvement.